

FAIRFORD, United Kingdom

WMO: 036440

Lat: 51.682N

Lon: 1.790W

Elev: 286

StdP: 14.54

Time Zone: 0.00 (EUW)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	26.6	21.0	15.9	27.4	23.4	17.9	29.8	31.5	49.3	28.7	49.7	3.7	250	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	79.7	64.8	75.6	63.5	72.9	62.1	67.6	74.9	65.5	71.9	63.8	69.7	8.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	94.8	69.1	63.3	87.9	67.0	61.9	83.5	65.7	32.1	75.4	30.5	72.5	29.2	70.4	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.5	23.0	20.2	19.4	85.7	6.0	4.0	15.1	88.5	11.5	90.9	8.1	93.1	3.7	96.0		
			20.0	70.6	4.3	2.6	16.9	72.4	14.4	74.0	12.0	75.4	8.8	77.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	39.5	40.7	44.1	48.2	54.1	59.1	63.2	61.4	57.7	52.5	45.0	40.5	(6)
	DBStd	9.79	6.19	5.74	5.02	5.28	4.92	4.49	4.73	3.80	4.47	4.86	5.85	6.98	(7)
	HDD50	1385	326	261	190	93	15	0	0	0	2	28	167	303	(8)
	HDD65	5327	789	680	648	503	340	184	92	122	221	387	600	761	(9)
	CDD50	1596	2	1	7	40	141	274	411	355	234	107	17	7	(10)
	CDD65	62	0	0	0	0	1	8	38	12	3	0	0	0	(11)
	CDH74	536	0	0	0	2	17	91	330	80	16	0	0	0	(12)
	CDH80	97	0	0	0	0	0	19	70	8	0	0	0	0	(13)
(14) Wind	WSAvg	9.1	10.0	9.9	10.2	8.9	9.5	8.8	8.7	8.7	8.2	8.5	8.7	9.6	(14)
(15) Precipitation	PrecAvg	30.6	3.2	2.2	2.0	2.2	2.4	2.2	2.2	2.5	2.2	3.0	3.3	3.1	(15)
	PrecMax	43.0	7.2	5.1	3.8	6.3	5.2	6.1	6.7	5.0	4.4	5.6	5.7	5.5	(16)
	PrecMin	21.7	0.4	0.3	0.5	0.1	0.3	0.4	0.3	0.1	0.6	1.1	1.2	0.8	(17)
	PrecStd	5.4	1.5	1.3	0.9	1.4	1.1	1.3	1.3	1.4	1.2	1.3	1.3	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	56.8	64.1	71.8	76.7	82.7	86.7	80.8	76.7	66.6	60.6	57.4	(19)
		MCWB	52.8	49.9	52.1	57.0	63.6	66.4	69.7	66.1	64.3	61.7	58.4	56.5	(20)
	2%	DB	53.3	53.0	58.7	66.4	71.6	75.7	80.9	75.6	71.9	64.1	57.6	55.4	(21)
		MCWB	52.2	50.3	52.4	56.0	60.4	63.4	64.9	64.1	63.3	59.9	55.9	55.0	(22)
	5%	DB	51.5	51.4	55.4	62.2	67.9	71.8	77.1	72.5	69.0	62.3	55.8	53.5	(23)
		MCWB	50.3	49.4	49.6	53.5	58.1	61.5	63.4	62.6	60.9	59.0	54.2	52.5	(24)
	10%	DB	49.7	49.8	52.6	58.6	64.0	68.4	73.5	69.8	66.5	60.5	54.2	51.5	(25)
		MCWB	48.7	48.2	48.3	52.1	56.1	59.9	62.0	61.5	59.9	57.6	52.6	50.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	53.3	55.5	59.9	65.0	68.9	70.9	68.7	69.6	63.6	59.3	56.7	(27)
		MCDB	54.2	54.1	57.4	67.9	72.2	79.5	82.1	75.1	72.4	65.6	59.7	56.8	(28)
	2%	WB	51.9	51.3	53.0	57.2	61.8	65.7	67.7	66.3	65.3	61.4	56.8	55.1	(29)
		MCDB	52.4	52.2	56.3	63.8	69.1	72.1	76.0	71.7	68.8	63.0	57.4	55.3	(30)
	5%	WB	50.1	50.0	51.0	55.1	59.4	63.0	65.4	64.6	63.1	59.8	55.1	52.8	(31)
		MCDB	50.7	50.8	54.0	60.1	65.2	69.3	73.9	69.7	66.7	61.4	55.9	53.3	(32)
	10%	WB	48.3	48.3	49.1	53.2	57.7	61.2	63.5	63.1	61.2	58.3	53.1	50.5	(33)
		MCDB	49.0	49.4	51.8	57.8	62.4	66.9	71.0	67.8	65.1	59.9	54.3	51.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.7	14.4	17.2	16.8	16.2	17.7	15.9	16.0	12.5	11.7	11.1	(35)
		MCDBR	11.2	13.1	17.8	23.8	22.5	21.7	23.3	19.7	18.8	13.1	11.9	10.4	(36)
	5% WB	MCWBR	11.4	10.2	12.3	13.8	12.3	11.3	10.2	9.8	11.0	9.4	10.7	9.8	(37)
		MCWBR	11.2	12.0	16.5	20.6	18.6	18.6	20.1	16.4	15.5	11.4	11.7	10.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.324	0.339	0.374	0.396	0.401	0.396	0.393	0.382	0.367	0.352	0.325	0.313	(40)
		taud	2.354	2.370	2.271	2.224	2.246	2.288	2.306	2.359	2.396	2.418	2.403	2.333	(41)
	Ebn at Noon		215	246	258	266	270	272	271	268	259	239	213	196	(42)
		Edh at Noon	20	26	34	40	41	40	38	35	30	24	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		252	464	810	1246	1522	1666	1575	1306	984	568	302	199	(44)
	RadStd		22	59	107	144	92	172	158	107	82	51	33	23	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (90 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page